

The CS Effect: Predicting Paper Success and Tracing the Spread of Computational Thinking

Text Analysis for Business

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Our research and why it matters



Generalizable features in citation prediction



Semantic drift of scientific disciplines

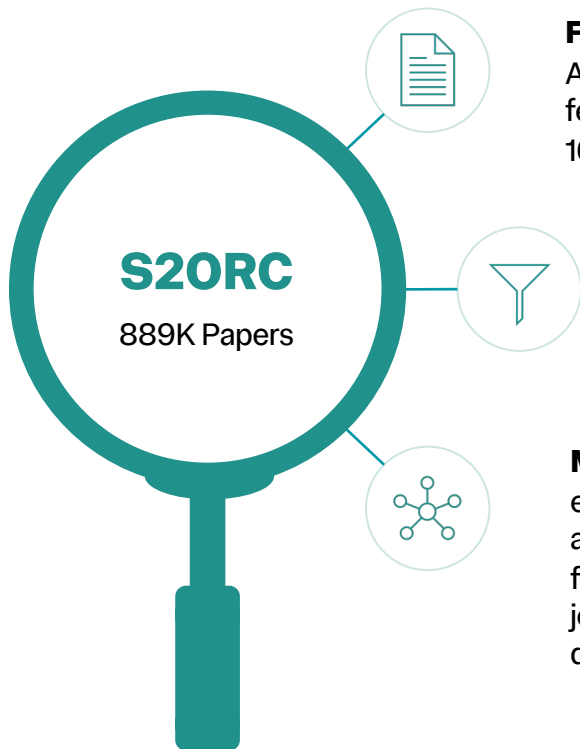


Transfer learning to other disciplines



Research gaps and opportunities

Dataset with 889K paper abstracts and relevant metadata



Focus on abstracts

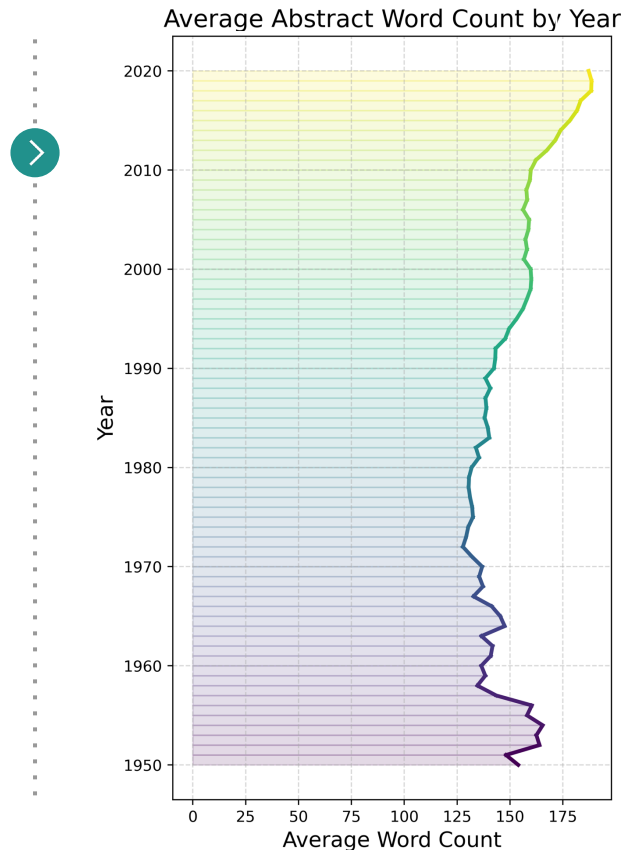
Analysis will focus on extracting textual features from the abstracts (average of 164.6 words)

Analysis-specific subsets

For each analysis a specific relevant sample is taken

Metadata

entities, s2Url, pdfUrls, s2PdfUrl, authors, inCitations, outCitations, fieldsOfStudy, year, venue, journalName, journalVolume, journalPages, sources, doi, doiUrl, pmid, magId



Understanding how CS “leakage” into other domains propels improvement in research quality

The hypothesis is that various fields have been strongly influenced by computer science and its applications. To quantify this influence, this research therefore focuses on identifying the commonalities in what makes papers successful from the perspective of computer science.



The proportion of AI-related research outputs over the different scientific fields exhibits a steady increase.¹



Studying how CS terminology and methodologies like AI “leak” into different domains can help determine if this digital convergence is driving higher research impact



Impact is measured by the number of citations, which signifies a paper serves as a foundation for other works and thus drives research in general forward

Transfer citation prediction from large corpus of CS papers to other domains

Research Question

How is computer science changing the other scientific fields and can this theme help us predicting paper citations?



Populations of interest

- Abstracts of CS papers in The Semantic Scholar Open Research Corpus
- Non-CS papers for transfer learning evaluation
- Field centroids in embedding space: the average semantic profile of each discipline



Quantities of interest

- Paper citation count
- Field loadings on semantic axes (PC scores)
- Inter-field semantic differences over time
- Cross-domain prediction transferability



Transfer Learning

Can the features learned exclusively from CS be used to predict outcomes for papers in other domains?

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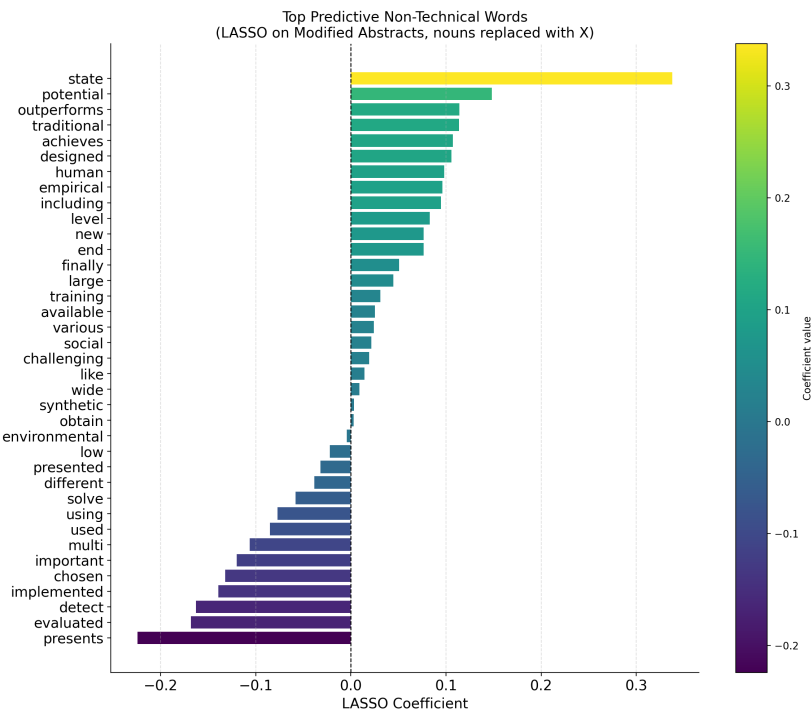
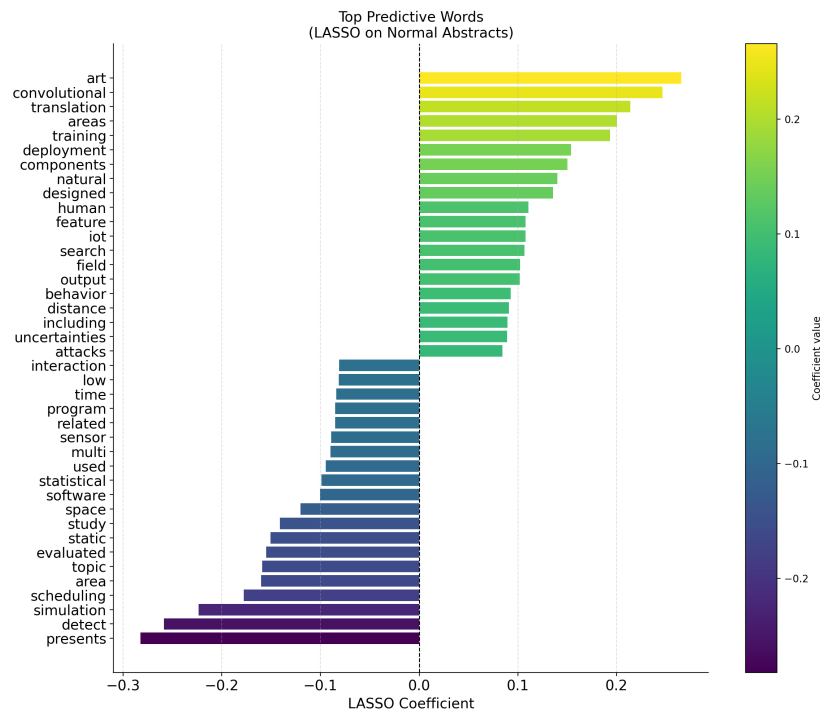


Transfer learning to other disciplines



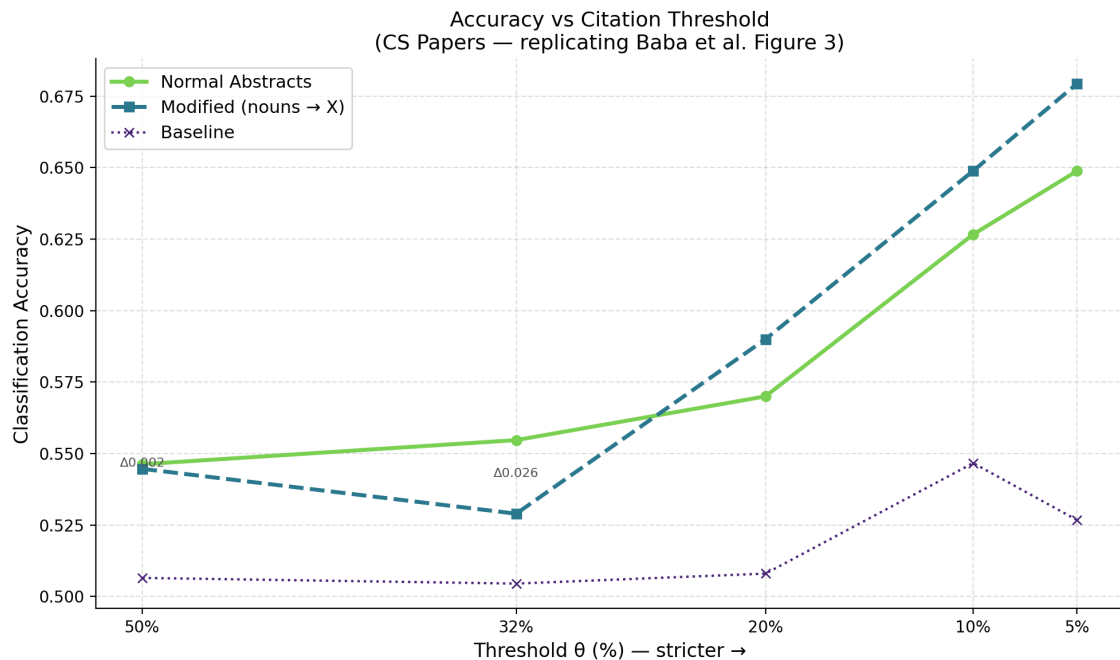
Research gaps and opportunities

Using a technique proposed by T. Baba, K. Baba and D. Ikeda², we find the non-technical terms in CS papers with prediction power



Replace technical terminology with X

Technical vocabulary does not appear crucial to distinguish between top and bottom performers in number of citations



The accuracy of citation-prediction models that do not hyperfocus on technical terms exceeds those of models trained on the untouched abstracts for the top and bottom percentiles of citation count ($\theta = 20\%$ or lower). This points to the presence of **generalizable patterns that can be informative when differentiating between high and low performers.**

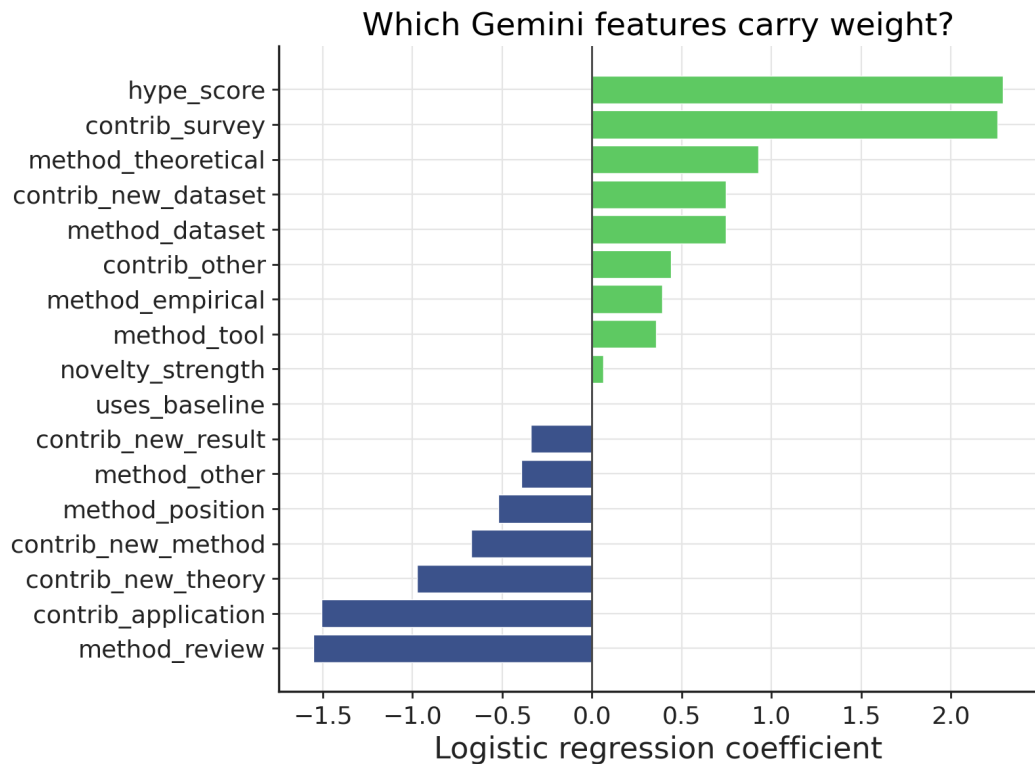
Although model performance is low overall, including the metadata delivers a slight improvement

Relying on textual data alone is not recommended, as **models that incorporate the available metadata seem to outperform** those only using the abstracts for both regression and classification with respective R^2 of 0.090 and accuracy of 0.595.

CS citation prediction — how much does each feature set add?



Other derived features may be useful in prediction



Gemini encoded relevant characteristics of the abstracts into 17 interpretable derived features. Feeding them into the same logistic regression reveals that two features carry almost all of the signal, a hype score and a survey indicator.

Interestingly, [a paper's claims about itself do not link back to the number of citations, but tone and genre do](#). Explicit claims of novelty or the presence of a comparison baseline are essentially irrelevant, as their coefficients are near zero.

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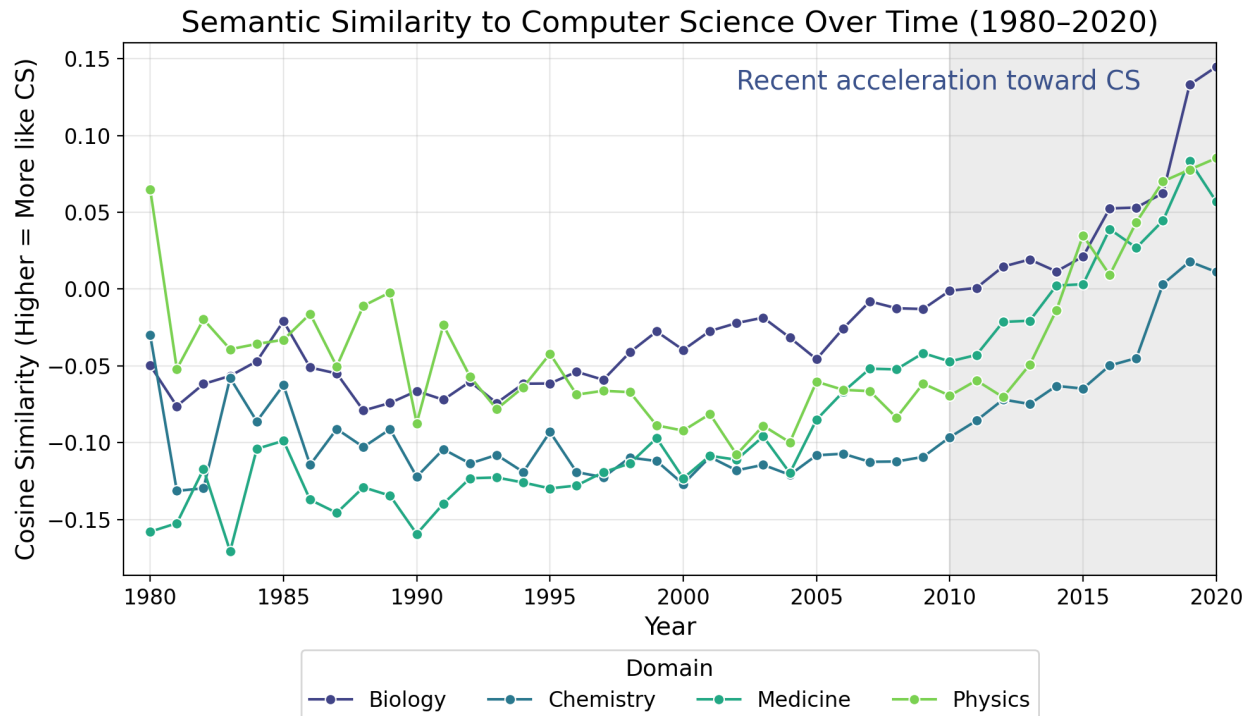


Transfer learning to other disciplines



Research gaps and opportunities

Embedding centroid cosine similarities show accelerated semantic drift towards computer science across domains from 2010

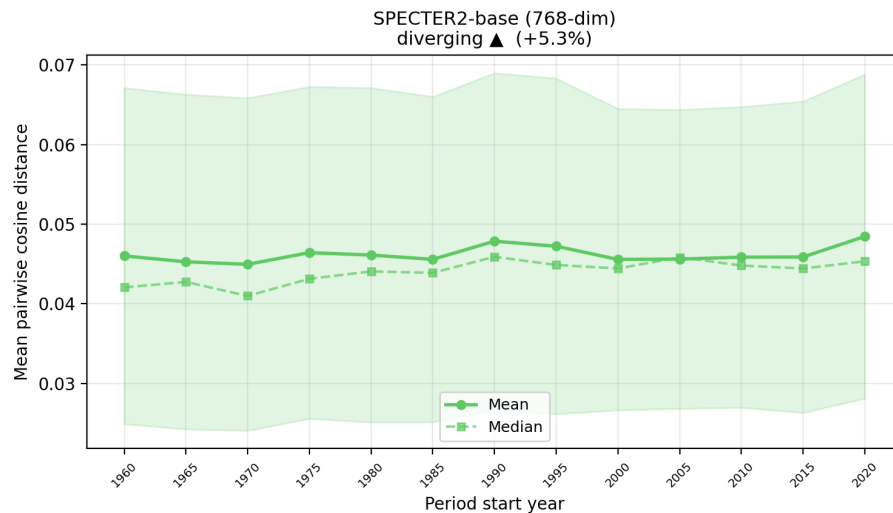
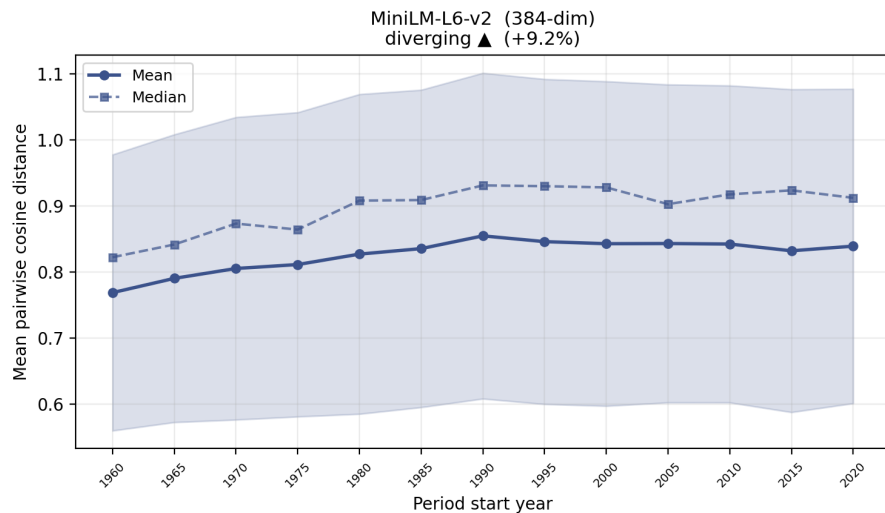


The cosine similarity between **the embedding centroids for each domain** and are noticeably **converging to CS**.

It is worthwhile to investigate how the embedding for CS itself and the different fields toward each other behave.

But scientific fields overall have become semantically more distinct over past 60 years

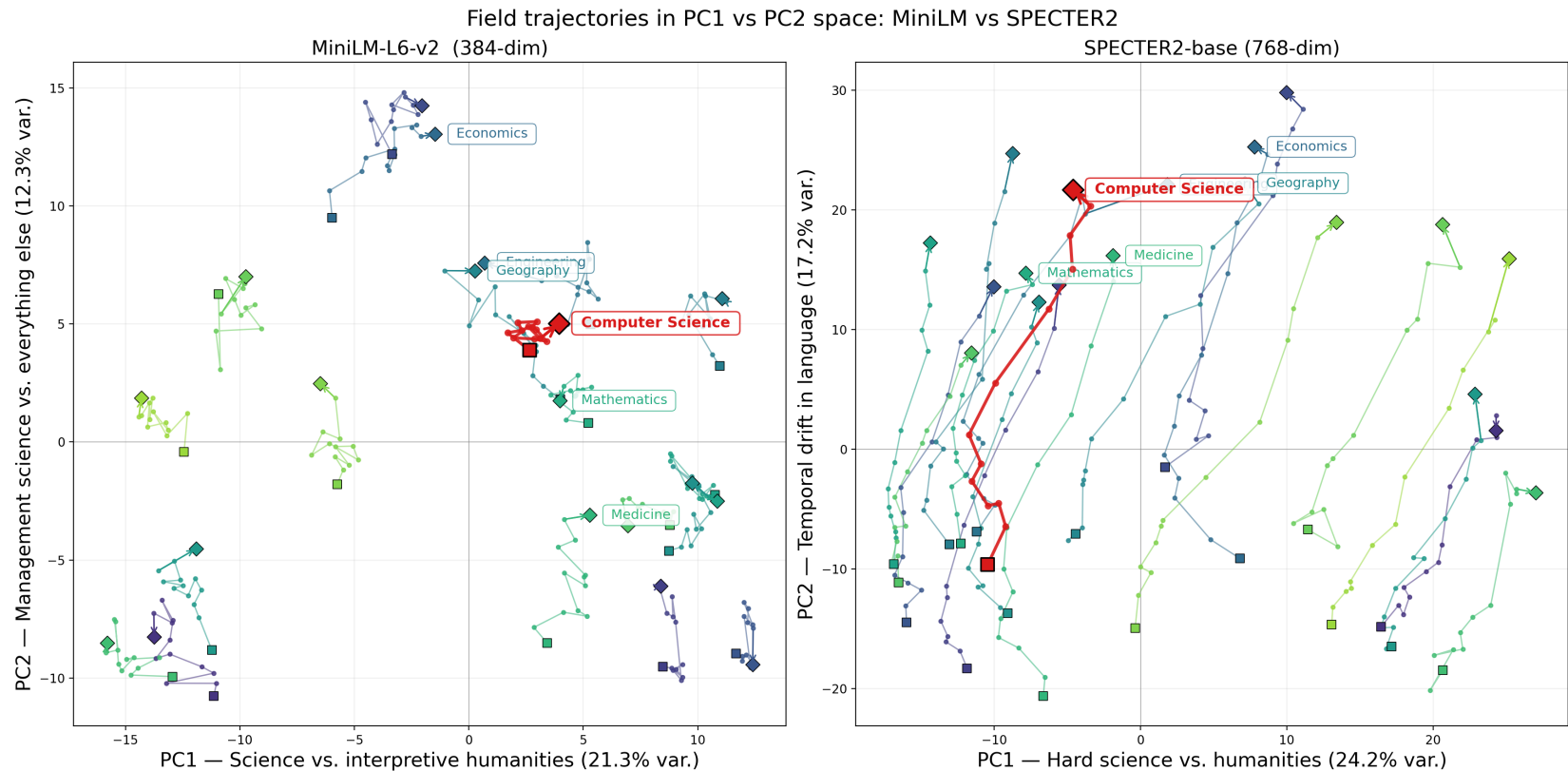
Field convergence over time: MiniLM vs SPECTER2
Do both models tell the same story?



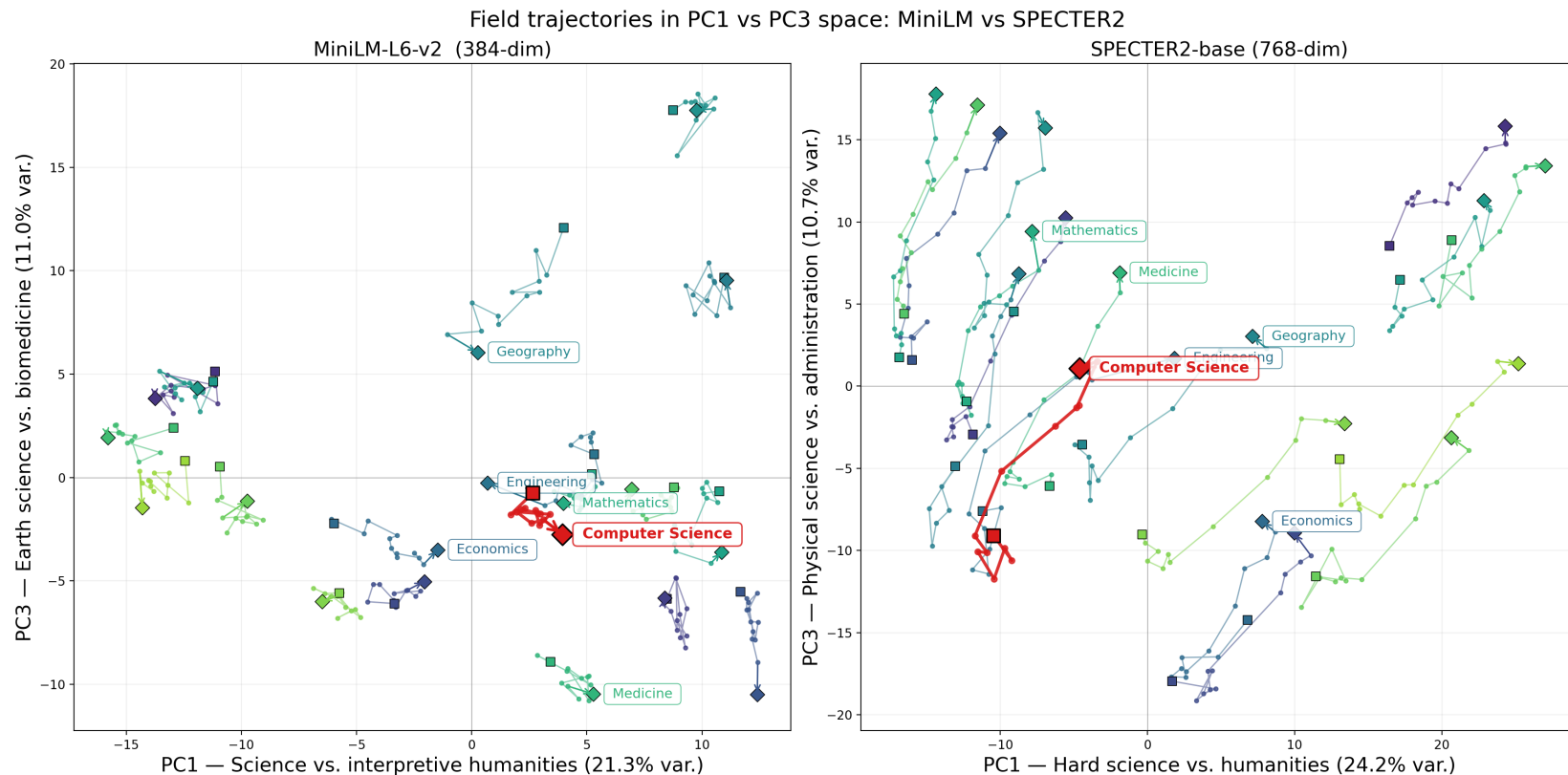
Hypothesis 1: Computer Science began as a semantically isolated field and converged toward the mainstream scientific cluster over time, while the cluster of mainstream sciences became more dispersed in terms of semantics.

Hypothesis 2: Most sciences moved further apart in most dimensions except the ones characteristic of computer science.

PCA on science-specific embeddings reveals significant semantic shift; papers are becoming more applied, and more field specific



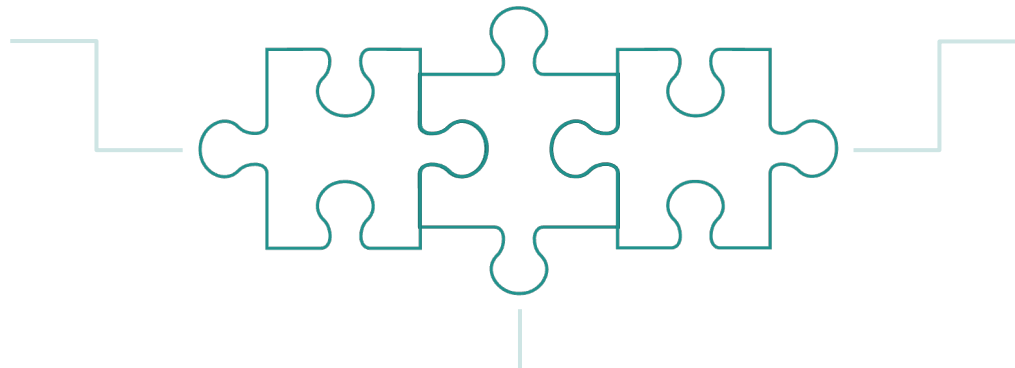
Field thematic identities were stable, but academic writing shifted from administrative toward empirical and quantitative vocabulary



Interpreting principal components required careful modelling choices since the dimensions reflected centroids and not actual abstracts

Extreme document inspection

Retrieved the 5 documents with the highest and lowest scores on each PC to identify what semantic content each axis captures qualitatively.



Metadata correlation

Computed point-biserial correlations between PC scores and field labels, and Pearson correlations with publication year, to validate whether the axes align with known disciplinary and temporal structure.

Vocabulary projection via Ridge regression

Trained a Ridge model to predict PC scores from TF-IDF token weights, then read off the top tokens at each pole (R^2 indicates how linearly recoverable each component is from document vocabulary)

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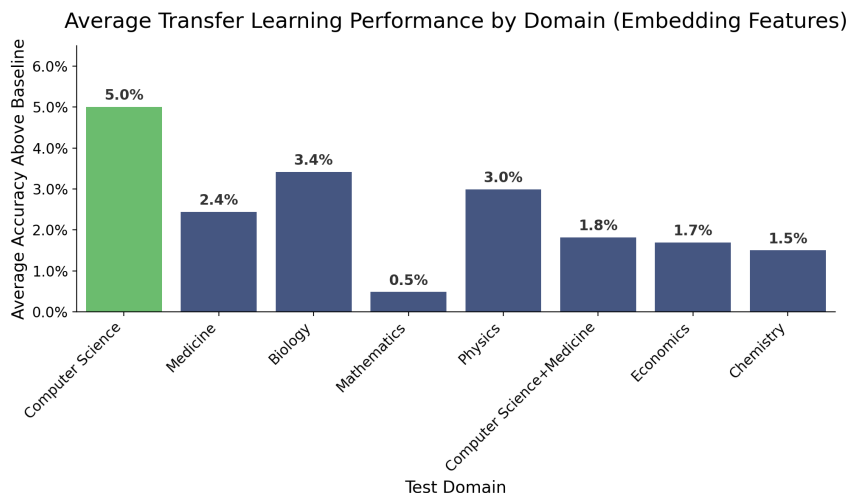
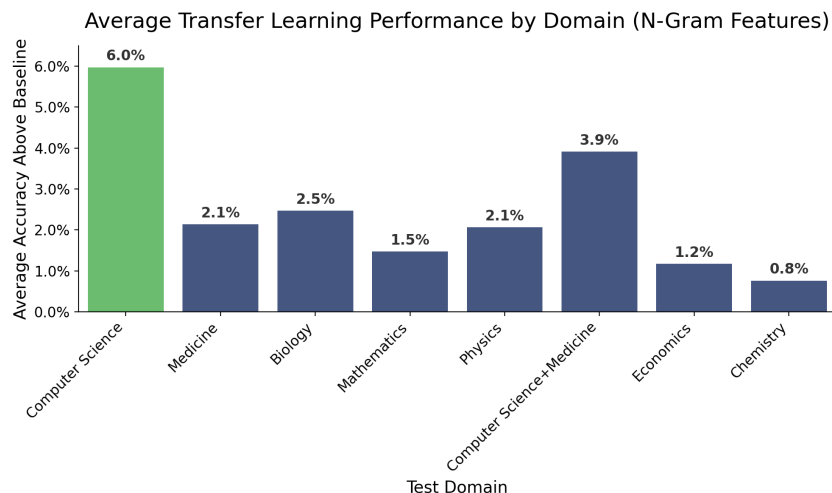


Transfer learning to other disciplines



Research gaps and opportunities

Transfer learning comparison for N-Gram and Embedding features show mixed results with overall signs of transferable knowledge



Average over multiple models trained on CS abstract features for different year segments starting from the 90's and tested on several domains:

- The CS-trained models were **consistently able to perform above the majority class baseline across most domains**, with varying performance for different domains.
- For N-Grams, the transferred performance was highest when compared to interdisciplinary papers (e.g. CS+Medicine), which makes sense since there is likely an overlap of words. However, this is not recognized for the embedding features.

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Unknown causal mechanism behind increasing similarity of other fields to CS



Convergence measured without controlling for within-field variance



Temporal vs. topical drift was not fully disentangled

1

Temporal vs. topical drift

Partial out the year effect by regressing PC scores on publication year and analysing residuals.

2

Centroid Sensitivity to Corpus Size

Weight centroids by inverse paper count or use median instead of mean

3

Within-field variance

Add interquartile range or standard deviation bands to convergence plots alongside the mean distance

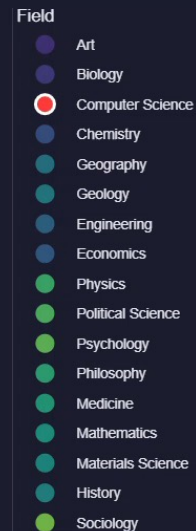
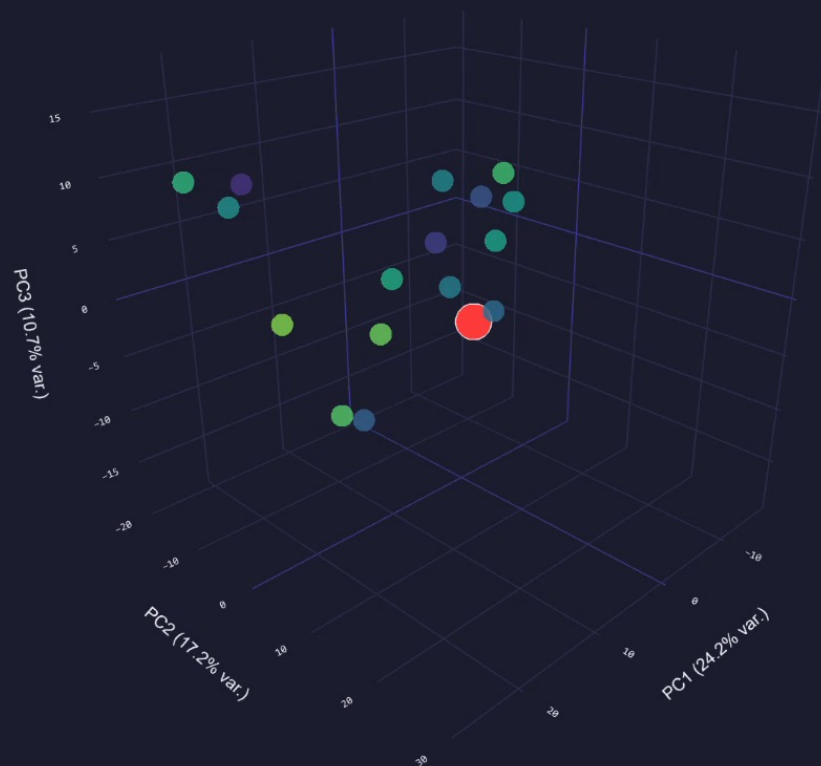
4

Post-2020 underrepresentation

Supplement with a more recent crawl from arXiv or Semantic Scholar.

Q&A





Period: 1960-1964



References

¹Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarlaschi, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, Carla Brodley, Jack Clark, Virginia Dignum, Vipin Kumar, James Landay, Terah Lyons, James Manyika, Juan Carlos Niebles, Yoav Shoham, Elham Tabassi, Russell Wald, Toby Walsh, Dan Weld. "The AI Index 2026 Annual Report," AI Index Steering Committee, Institute for Human-Centered AI, Stanford University, Stanford, CA, April 2026.

²T. Baba, K. Baba and D. Ikeda, "Citation Count Prediction Using Abstracts," in Journal of Web Engineering, vol. 18, no. 1-3, pp. 207-228, January 2019, doi: 10.13052/jwe1540-9589.18136.